



Make Your Own Password Locker

Fayaz Hassan

A password locker usually has an electromagnetic lock that opens only when the correct password is entered. The circuit presented here for such a locker is a simple, low-cost, hi-security six-digit password-operated system. ATtiny2313 microcontroller (MCU) used in the circuit is easily available and programmable using any AVR programmer.

The circuit diagram of the password locker is shown in Fig. 1. It is built around ATtiny2313 (IC1), 2N2222 transistor (T1), 16x1 LCD (LCD1), 5V single-changeover relay and a few other components.

The twenty-pin ATtiny2313 MCU is the heart of the project. First, a six-digit password has to be saved in the EEPROM of the MCU. The lock opens when this password is entered. The saved password is available even after disconnecting the power supply. But the six-digit password can be erased,

and a new password generated, if required.

Only three buttons are used here for entering or saving the password. A blinker LED (LED1) is used as the status indicator of the MCU process.

One out of the six digits of the password is displayed at a time and allowed to be edited. Remaining five digits appear as asterisks (*) in the LCD display. Cursor of LCD1 blinks when password digits are entered.

Up button is used to change the digit from 0 to 9. Similarly, Right button is used to move the cursor right for editing. Once the password is entered, press Enter to start processing the entered password.

Jumper J1 is provided to erase the existing password. Close J1 and then power on the circuit; the existing password, if any, will be erased and the new password may now be entered. Once a new password is successfully saved to EEPROM,

disconnect or open J1, which should not be accessible to any other user.

In case of a new chip, while using

PARTS LIST

Semiconductors:

IC1	- ATtiny2313 MCU
T1	- 2N2222 transistor
D1	- 1N4007 rectifier diode
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 10-kilo-ohm
R2-R5, R8	- 4.7-kilo-ohm
R6	- 2.2-kilo-ohm
R7	- 2.7-kilo-ohm
R9	- 47-kilo-ohm
R10	- 1-kilo-ohm
VR1	- 4.7-kilo-ohm preset

Capacitors:

C1, C2	- 100nF ceramic disk
--------	----------------------

Miscellaneous:

CON1	- 2-pin terminal connector
J1	- 2-pin Berg male connector
S1-S3	- Tactile switch
RL1	- 5V, single-changeover relay
LCD1	- 16x1 LCD
	- Shorting jumper cap for J1
	- 20-pin IC base

(All these are available from kitsnspares.com)

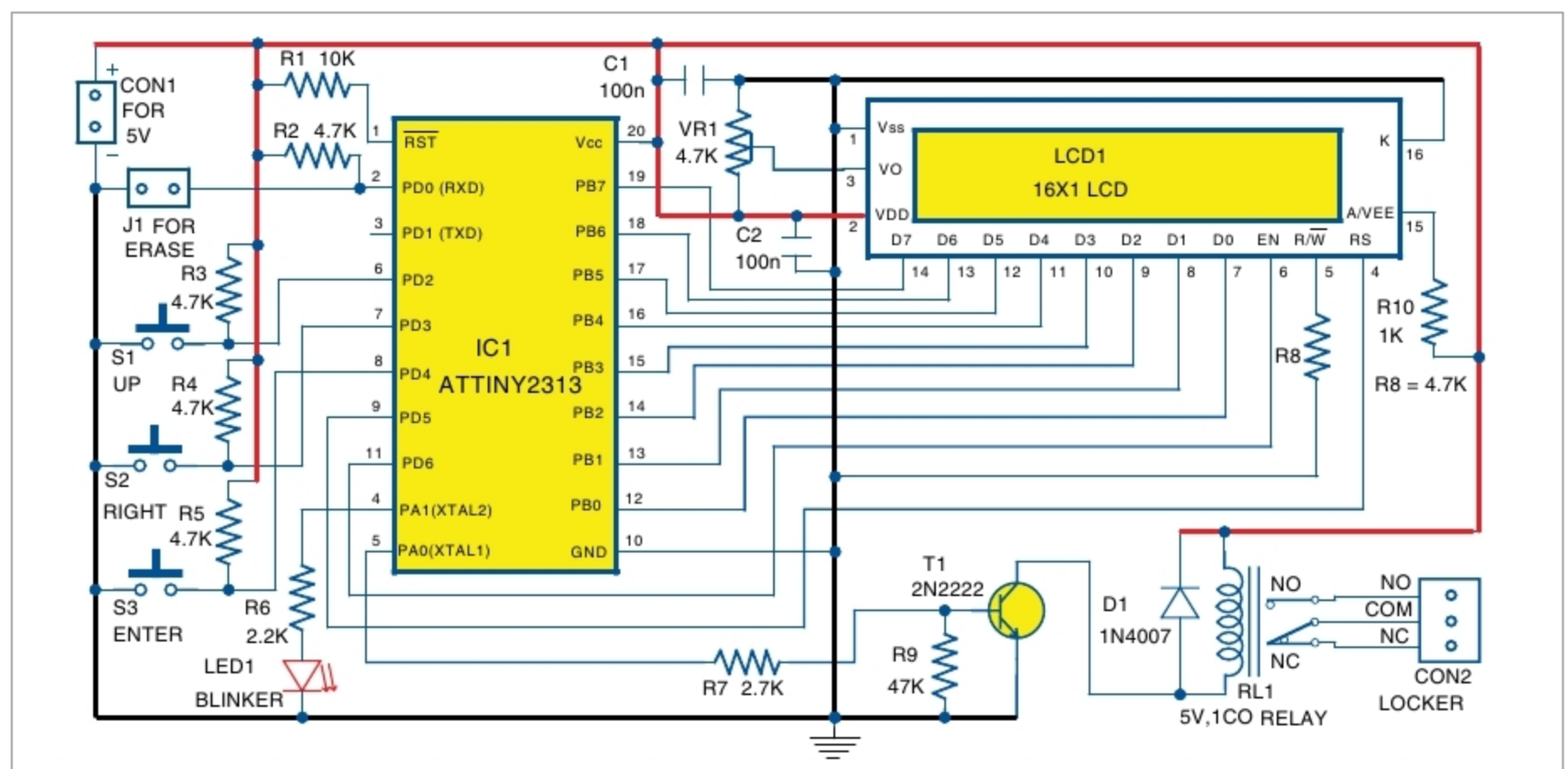


Fig. 1: Circuit diagram of the password locker